

Umer Niazi

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EDUCATION

University of Engineering and Technology, Lahore

BS in Artificial Intelligence

Lahore, Pakistan

2025 – 2029 (*Expected*)

Forman Christian College

Intermediate in Computer Science (ICS)

Lahore, Pakistan

2022 – 2024

PROJECTS

News NLP Pipeline | *Python, BERTopic, spaCy, Transformers, Streamlit, SQLite*

May – July 2026

- Built an end-to-end NLP pipeline for unsupervised topic discovery, sentiment analysis, and named-entity extraction across 15 years of Dawn News headlines without manual labeling
- Scraped and preprocessed 350K+ headlines; modeled semantic representations using BERTopic with Transformer embeddings to extract 20 distinct topic clusters
- Extracted 420K+ named-entity mentions and deployed an interactive Streamlit dashboard for exploring topics, sentiment trends, and entities

Himalayan Wildlife Classifier | *Python, PyTorch, ResNet18, Streamlit*

May – July 2026

- Built an image classifier for Snow Leopard, Markhor, and Himalayan Brown Bear, plus a background Other class, using transfer learning on a frozen ResNet18 backbone
- Achieved 96.55% validation accuracy; implemented an 80% confidence threshold for selective prediction to filter ambiguous and low-confidence inputs
- Deployed an interactive Streamlit interface for real-time inference with per-class confidence scores

wmus | *Python, windows-curses, pygame*

Jan – June 2026

- Built a native Windows terminal music player inspired by the cmus workflow, featuring keyboard-first navigation and a custom curses interface
- Implemented fuzzy track search, metadata caching for fast library startup across large audio collections, and fully configurable keybindings
- Packaged as a standalone Windows installer using PyInstaller and Inno Setup, eliminating the need for a local Python runtime

TECHNICAL SKILLS

Languages: Python, C++, GDScript

Machine Learning & NLP: PyTorch, Transformers, scikit-learn, Hugging Face, spaCy, BERTopic, NLTK

Data: Pandas, NumPy, SQLite

Systems & Tools: Linux, Bash, Git

Development: Streamlit, Godot, Unity